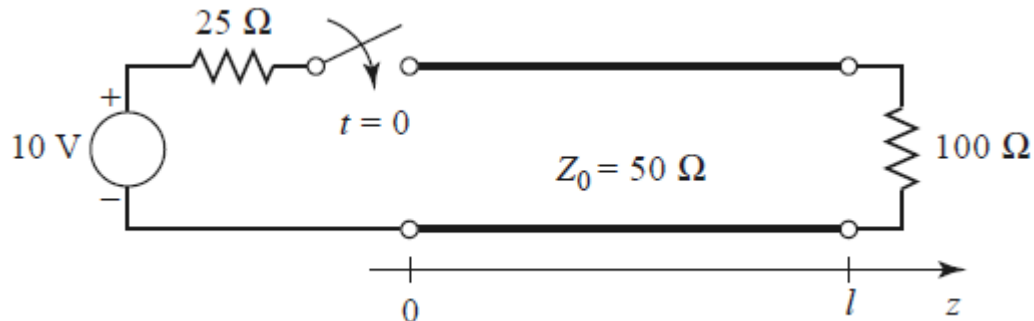


EE 340 Electromagnetic Theory

Assignment #04

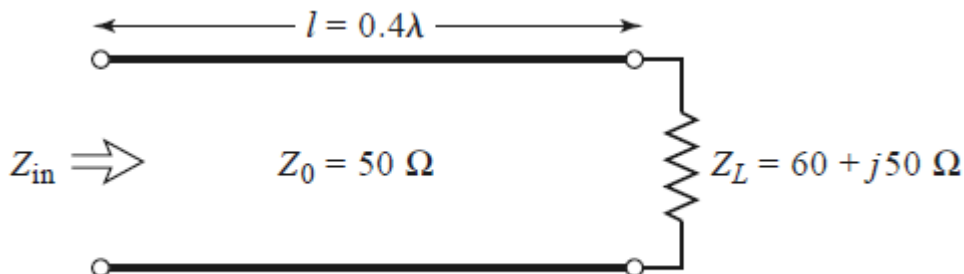
Q.1 Plot the bounce diagram for the transient circuit shown in the accompanying figure. Include at least three reflections. What is the total voltage at the midpoint of the line ($z = l/2$), at time $t = 3l/v_p$.



/vp

Q.2 Use the Smith chart to find the following quantities for the transmission line circuit shown in the accompanying figure:

- (a) The SWR on the line.
- (b) The reflection coefficient at the load.
- (c) The load admittance.
- (d) The input impedance of the line.
- (e) The distance from the load to the first voltage minimum.
- (f) The distance from the load to the first voltage maximum.



Q.3 Use the Smith chart to find the shortest lengths of a short-circuited 75Ω line to give the following input impedance:

- (a) $Z_{in} = 0$
- (b) $Z_{in} = \infty$
- (c) $Z_{in} = j75\Omega$
- (d) $Z_{in} = -j50\Omega$
- (e) $Z_{in} = j10\Omega$

Q.4 Consider two two-port networks with individual scattering matrices $[S^A]$ and $[S^B]$. Show that the overall S_{21} parameter of the cascade of these networks is given by:

$$S_{21} = \frac{S_{21}^A S_{21}^B}{1 - S_{11}^B S_{22}^A}$$

Q.5 A load impedance $Z_L = 100 + j80\Omega$ is to be matched to a 75Ω line using a single shunt-stub tuner. Find two designs using short-circuited stubs. Use smith chart.

Q.6 A load impedance $Z_L = 90 + j60\Omega$ is to be matched to a 75Ω line using a single shunt-stub tuner. Find two designs using open-circuited stubs. Use smith chart.

Q.7 Write a MATLAB code to design a single shunt-stub tuner.